LIBBY — PATIENT/CAREGIVER TRACK

ulms-highgrade-pulm-mets-tp53-smarcb1-hla-a2-w9t4

Plain-language summary for patient and caregiver

Generated 2026-08-25

What this page is

Libby is a research tool. It takes the specific features of one person's cancer, gathers the treatments aimed at those features, and lays out what the published evidence says about each one. This page is the plain-language version of that work for your case: what is known about your cancer, what has not been measured yet, the planned chemotherapy with its real numbers attached, and then the experimental options in order, with the upside and the catch of each written as plainly as we can manage.

One thing to hold onto before anything else: this analysis produced two lists, and each one numbers from 1. The standard-of-care list holds the approved, guideline-backed treatments for this cancer, and the chemotherapy your oncologist has already planned sits at the front of it. The experimental list holds 45 entries, trials and off-label ideas aimed at your tumour's specific features, and the first entry on it is a set of tests rather than a treatment. Number 2 on one list has nothing to do with number 2 on the other. On this page I mostly use treatment names instead of numbers, and where a number does appear I say which list it belongs to. The full standard-of-care page for your case is still being written, so the chemotherapy section below is drawn directly from the trial that tested it.

What we know about your cancer

You have leiomyosarcoma, a cancer that starts in smooth muscle, in your case the muscle wall of the uterus. It is high grade, meaning fast-growing under the microscope, and it has spread: there are multiple small tumours in both lungs. A sarcoma specialist read your scans as showing an unusual, spread-out pattern rather than neat round spots, and that detail comes back later because it affects one of the surgical options.

The mass removed in your surgery in September 2024 was called benign at the time. Re-review corrected that in July 2026, roughly 23 months later. That delay was real, it was not your doing, and you should not have had to wait that long for the truth. Two facts come out of it that your doctors can actually use. You have never had chemotherapy, so every first-line option is still open. And because nothing was treating the disease in the interval, its roughly two-year pace is visible, which matters for one of the surgery-based options below.

Beyond that, the plain truth is how little has actually been measured.

- The hormone receptor tests came back essentially negative (under 1 percent). That is a real answer: it closes the door on hormone-based treatments, which only work when the receptors are present.
- A blood test, a "liquid biopsy" that looks for tumour DNA in the bloodstream, reported a change in a gene called SMARCB1. It looked promising at first, and you may have been told it was important. A section further down deals with it honestly: it is very likely a false lead.
- The same blood test reported a change in a gene called TP53. There is a catch: TP53 changes on blood tests can come from ordinary age-related changes in blood cells rather than from the tumour, and only a tissue test settles which. Even if it is from the tumour, it is expected in this disease and no treatment on this page targets it directly.
- Your tissue type (HLA) was tested, and it includes a type that some engineered immune-cell therapies require. That opens a door partway. Those therapies also need the tumour itself to display a particular marker, and none of those marker tests have been done. The typing alone does not qualify you for anything.
- Almost everything else that would steer treatment choice has never been tested on the tumour itself. The stains and gene panels that decide most of the experimental list are all still pending, and the blood test cannot stand in for them.

- No baseline organ tests are on file: no heart scan, no lung function tests, no oxygen reading, no blood counts. All of them are needed before the planned chemotherapy anyway.
- Your day-to-day fitness score (doctors call it ECOG performance status) was assumed to be 1, meaning up and about with some symptoms. Nobody has actually assessed it. Every study result quoted on this page came from patients with a measured score of 0 or 1, so this is not paperwork; it is part of whether those results apply to you at all.
- You live at roughly 8,500 feet and have never smoked. Altitude changes what a normal oxygen reading and normal blood counts look like for you, and several trials set oxygen requirements written for sea level. Nobody has accounted for this yet.

What you told us matters most

The honest answer is that nobody asked, and this page should not pretend otherwise. The preferences on file for your case are defaults: a lean toward effectiveness over gentleness, no side effects ruled out, no travel limits, open to clinical trials. Not one of those came from you. What you are aiming for, what you would give up, how far you would travel, which risks you would refuse: none of it has been discussed. A woman in her thirties heading into aggressive first-line treatment is usually assumed to want the most disease control available, and that assumption may well be right, but it is an assumption. It should be replaced with your actual answers, and there is a question about exactly that at the bottom of this page.

The treatment to talk about first: the planned chemotherapy

The plan your sarcoma oncologist has already proposed is doxorubicin plus trabectedin, given together for six cycles, then trabectedin alone as maintenance. This regimen is the lead recommendation of this entire analysis, ahead of everything on the experimental list. It belongs to the standard-of-care list, and it is the only treatment anywhere in your case with a survival number behind it.

That number comes from a randomised trial of 150 people with leiomyosarcoma who had never had chemotherapy, a little under half of them with uterine disease. Half received the combination and half received doxorubicin alone, the standard at the time. People on the combination lived a median of about 33 months, compared with about 24 months on doxorubicin alone, and the cancer was held in check for a median of about 12 months compared with about 6. In a disease where randomised survival gains are rare, that is the strongest result on record.

Now the part nobody has put to you yet, and it should be said in numbers before you sign anything. The combination is much harder on the bone marrow than doxorubicin alone. In the trial, about 8 in 10 people on it had severe drops in their infection-fighting white cells. About 28 in 100 developed a fever while their white cells were down, and that is not a nuisance side effect: it usually means being admitted to the hospital for IV antibiotics. Nearly half had severe platelet drops, and about 3 in 10 had severe anaemia. One person in the trial died of a treatment-related heart problem, and that was in the doxorubicin-alone group, not the combination group. Most of this is managed with dose adjustments and supportive care, and the trial reported no treatment-related deaths on the combination, but "managed" is a word doctors use. You are the one who will live it. You are owed this conversation, with these numbers in it, before you agree.

To be clear: nothing in this analysis argues against this chemotherapy. The recommendation is to start it with open eyes, with the tests below running alongside it, and with the dose decisions in the next section made in advance rather than discovered later.

The first step everyone agreed on

The first entry on the experimental list is not a drug. It is one tissue request and one clinic visit, and it comes first because almost nothing about your tumour has been measured and nearly every other decision in your case waits on the results.

Your tumour tissue exists in two places: the 2024 uterine specimen, held at the hospital that did your surgery, and the 2026 lung biopsy. Nearly every pending test runs on those blocks. The request should go out as a single order naming both specimens and covering everything at once:

- a stain for a protein called INI1, which settles whether the SMARCB1 blood-test finding is real (more on that below);
- a stain for a protein called Rb, which decides which of two trials fits you (also below);
- a full gene panel on the tumour itself, including a DNA-repair-deficiency score that would strengthen the case for the olaparib option, and which also settles the TP53 question;
- stains for three immune-therapy markers (MAGE-A4, PRAME, NY-ESO-1), the missing second half of the cell-therapy door your tissue type opened;
- a few further markers that ride along on the same tissue cut at almost no extra cost.

The clinic half: an actually measured fitness score, a heart scan (required before doxorubicin anyway), blood work, lung function tests, and an oxygen reading with your home altitude written on the report. That last detail is small and it matters. Some trials in this file require oxygen levels of at least 90 percent, one of them 92, and a reading that looks borderline to a screening coordinator at sea level may simply be your normal at 8,500 feet. If the altitude is not on the report, nobody far away can know that.

None of this is treatment and none of it is toxic. It is stains on tissue that already exists, a clinic visit, a heart scan, a blood draw. The stains take about 3 to 5 days once a block is in the lab; the gene panel takes 2 to 3 weeks. The slow part is not the lab, it is the paperwork: getting a 2024 block released from an outside hospital, the same one that originally called the tumour benign, is the kind of request that sits in a queue for weeks unless a named person owns it and chases it. Right now nobody does. Ordering these tests one at a time instead of on one request is how weeks get lost.

This step runs alongside starting chemotherapy, never instead of it; nothing here justifies delay. One cost to know about: the immune-marker stains are often not covered by insurance outside a trial, so it is worth asking the self-pay price and asking trial sponsors whether they run their own version free during screening.

Decisions that belong before the first cycle

Three of them. All cheap now, expensive later.

A stopping rule for doxorubicin. Doxorubicin has a lifetime dose ceiling because of what it can do to the heart, and two options you might want later have their own stricter ceilings written into their entry rules. The Toronto lung-perfusion study excludes anyone whose total doxorubicin dose passed 450 mg. A separate drug trial further down the list caps the dose at 360 mg per square metre of body surface, and six full cycles of the planned regimen lands exactly on that line. This is arithmetic, not judgment. Whether to stop at five cycles, go to six, or plan differently is a decision that costs nothing made before cycle one, and may cost two open doors if it is first noticed during cycle six.

The gemcitabine trap. The Houston trial described below permanently excludes anyone who has ever received the chemotherapy drug gemcitabine. The planned regimen contains no gemcitabine, so it keeps that seat open. But gemcitabine plus docetaxel is a common fallback in this disease, and one off-protocol prescription of it would close the Houston door for good. There is one sensible exception already thought through: if the heart or lung workup rules out full-dose doxorubicin and the trial's screening timeline cannot beat your symptoms, gemcitabine-based treatment close to home is a defensible choice. It should be made knowingly, not by default.

Starting one drug can close another trial. The New York study described below excludes anyone who has previously had an immune-checkpoint drug or a blood-vessel-pathway drug such as pazopanib. If that study is ever going to be considered seriously, the considering has to happen before those drugs are used.

The experimental options

Five entries on the experimental list were ranked as live options behind the workup. They appear here as Options 1 to 5, in that order. These option numbers belong to this page; the clinician-facing experimental table lists the same five as its ranks 2 through 6, behind the workup at its rank 1, and the standard-of-care list is numbered separately again.

Option 1 — Olaparib plus temozolomide, for after the first line

This is the strongest experimental signal in your exact disease. Olaparib blocks a DNA-repair enzyme (a PARP inhibitor); temozolomide is an oral chemotherapy that damages tumour DNA; the pairing is designed so the damage cannot be repaired. In a 22-woman study in advanced uterine leiomyosarcoma, all previously treated, tumours shrank by formal criteria in 6 of 22, roughly 27 in 100, and the typical time before the cancer grew again was about seven months. For a previously treated group with this disease those are real numbers, and the study met the goal it set for itself in advance, which matters because small studies often move the goalposts. A larger randomised study has finished enrolling and has not yet reported; it will either confirm this result or deflate it.

The caveats. The study had no comparison group, so nobody can say exactly how much of that seven months the drugs earned. It enrolled people after at least one prior treatment, so this is not a rival to the planned chemotherapy; it is a candidate for afterwards. Both trials of the combination are closed, so the route would be an off-label prescription of two approved drugs, and insurance approval is far easier to win with a tissue repair-deficiency result in hand, which is exactly why that test sits in the first step now rather than later. And it is hard on the marrow: about 75 in 100 had severe white-cell drops and about a third had severe platelet drops, landing on a marrow that would just have absorbed the planned regimen's similar rates. Two marrow-punishing regimens back to back is not the same proposition as either alone, and that sequencing question deserves its own conversation when the time comes.

One number to be wary of if you read the paper: a subgroup with a particular repair-test result appeared to do about twice as well. That analysis was exploratory, in 18 people. It is a hypothesis, not a result, and nobody should quote it to you as a prediction.

Option 2 — Abemaciclib added to gemcitabine and docetaxel, in a randomised trial in Houston

A single-centre trial is testing whether adding abemaciclib, a pill that blocks cell-division signals, improves on gemcitabine plus docetaxel, a standard chemotherapy pairing in this disease. What makes this seat unusual on this page: it accepts patients who have not yet had chemotherapy, the planned regimen does not disqualify you from it later, and if you enrol and land in the comparison group, what you receive is standard, guideline-supported chemotherapy. Its floor, in other words, is ordinary care. No other trial here can say that.

Be equally clear about what is unknown. Abemaciclib has essentially no track record in leiomyosarcoma, there is no efficacy number for the addition, and part of the trial's purpose is to establish that the combination is safe at all, including a fair open question about whether a drug that pauses cell division might blunt a chemotherapy that attacks dividing cells. The chemotherapy backbone has its own record: in a study of 42 women with untreated uterine leiomyosarcoma, about 36 in 100 responded, disease control lasted a median of around four and a half months, and one of the 42 had a severe lung complication possibly related to treatment, a number worth weighing for someone with disease in both lungs living at altitude.

Entry has two gates. It requires the tumour to have kept a protein called Rb, and that stain, part of the first step, has not been done. The same stain, if it shows Rb lost instead, routes to a different early-phase trial at the same centre. And the trial lives in Houston, which for you means travel during treatment; whether that is acceptable is a question nobody has asked you. Remember the trap above: any gemcitabine outside the trial closes this door permanently.

Option 3 — IMA203, immune cells engineered against a marker called PRAME

This option carries caveats. Here is the tradeoff to weigh.

This is the one cell-therapy door your tissue-type result actually opens. IMA203 takes your own immune cells, engineers them to recognise a marker called PRAME that some tumours display, multiplies them, and returns them after a short course of chemotherapy that makes room for them. It requires exactly the tissue type you have. It also requires the tumour to display PRAME, which has never been tested, and the company uses its own in-house test to decide, so even the stain in the first step cannot settle entry by itself.

What the evidence shows: in an early-phase study of 41 heavily pre-treated patients, mostly melanoma and none with uterine leiomyosarcoma, about 29 in 100 responded and the typical response lasted about four and a half months. Two of the 41 had a severe form of the immune overreaction called cytokine release syndrome. Those numbers come from other cancers and later lines of treatment, and they do not transfer to your situation; for your case the only fair label is unknown. The study's rules also require standard treatment first in practice, so like Option 1 this is an after-chemotherapy candidate.

What is contested here is the reason for wanting it. Chasing an early-phase cell therapy for the chance of a lasting response is a preference, and it has not been asked of you. The option stays on the page because the useful action right now costs nothing: the sponsor's screening conversation takes weeks and runs alongside chemotherapy. One question should be put to them directly, because their study listing names some uterine cancers and not this one: is uterine leiomyosarcoma accepted at all?

Option 4 — Chemotherapy washed through the lungs during surgery, then removal of the tumours (Toronto)

This option carries caveats. It is also the only option in your entire case aimed at cure rather than control.

The background fact is that surgeons do sometimes remove lung metastases in sarcoma, and in carefully selected patients the registry numbers are real: in one series, over half of the selected uterine-sarcoma patients were alive at five years; in a larger international registry, about a third after complete removal. Selected is the load-bearing word. Those patients had countable nodules that could all be removed, and slower-paced disease, typically more than a year between first surgery and spread. Your 23-month interval fits that profile comfortably. Your current scans do not: multiple tumours in both lungs, with that unusual spread-out pattern. Whether your disease ever becomes operable depends on how well the chemotherapy works, and that call belongs to a thoracic surgeon looking at your actual films, not to anyone at a distance.

The Toronto study adds an experimental step: during the operation, the lung's blood supply is isolated and washed with doxorubicin, aiming at microscopic disease the knife cannot see. That technique is at the earliest human stage. Its own primary question is whether it injures the lung, and no efficacy result for it has been published in anyone. The concern worth stating plainly: an early-stage surgical technique, in a person whose lung reserve has never been measured, living at altitude, with an infiltrative pattern on scans, stacks several unknowns on top of each other.

Two questions can be asked this week without committing to anything: does the study accept US residents, and how does its doxorubicin ceiling count a planned six-cycle induction? The second is why the stopping rule above exists. And removing lung metastases without the experimental wash is standard practice in selected patients; it lives on the standard-of-care list and stays available regardless of what happens with Toronto.

Option 5 — Ivonescimab, a two-in-one immune and blood-vessel drug (New York)

This option carries caveats, and they are heavier than any other ranked option here.

Ivonescimab is a single drug built to do two jobs: release an immune brake (the PD-1 pathway) and block a blood-vessel growth signal (VEGF). In lung cancer it beat a standard immune drug head-to-head, holding disease in check for a median of about 11 months versus about 6. That is an impressive result, and it has nothing to do with sarcoma. In your disease, both halves of the mechanism have already been tested separately and did not work: a blood-vessel-blocking antibody added nothing to first-line chemotherapy in a randomised trial in this exact cancer, and immune-checkpoint drugs produced zero responses across two studies in this disease, one of which was stopped early for futility. The hope in the new drug is that the two halves together achieve what neither did alone. That idea has laboratory support and, so far, no clinical support in any sarcoma.

The reason it is on the page at all is access. The New York study is the one trial here whose written rules would accept you right now, before chemotherapy starts, because patients who decline standard first-line treatment are eligible. That is a real fact and a dangerous one, because using that access means spending your first-line window, the only period in which the 33-month regimen is on offer, on two mechanisms this disease has already beaten. Most of the evidence weighs against this option, and it sits last among the ranked five for that reason. It stays visible because whether you would even want it is a question that has never been put to you. Practicalities, if it were ever pursued: sites are in the New York and New Jersey area, travel is not paid for by the study, its known side effects include high blood pressure and bleeding risk (worth weighing with tumours in both lungs), and the door closes permanently if a checkpoint or blood-vessel drug is ever used first.

Further down the list

Thirty-nine more entries sit on the experimental list below the ranked five. Most carry a flag: too little evidence, not open to someone in your situation, or no longer available. They stay on the list so you can see what was weighed, not only what survived. A few deserve plain words.

The SMARCB1 finding, and the drug built for it. The blood test found a damaging change in a gene called SMARCB1, and a drug called tazemetostat exists for cancers that have fully lost this gene's protein; it is approved in a different, rare sarcoma. **That option is only on the table if the INI1 stain in the first step shows the protein missing from the tumour, and it very probably will not.** Here is the honest picture, because you may have been told this finding was important. In the one published series that checked, 170 out of 170 uterine smooth-muscle tumours had kept this protein. The blood-test call has not been confirmed in the tumour itself, and blood-only findings sometimes are not from the tumour at all. And the change was found in only one of the gene's two copies, while the treatment is built for tumours in which the protein is fully gone, a different pattern from the one your result shows. A weak, unconfirmed signal like this is a much less sure sign that a treatment would help than a confirmed full loss would be, so this sits well below the options above it. The stain settles it either way in under a week. If it comes back normal, this branch closes: the SMARCB1-directed entries come off the experimental list, and everything else on this page, including the chemotherapy plan on the standard-of-care list, stands exactly as written. Even in the unlikely event the stain confirmed a loss, the drug's record in such tumours outside its approved disease is about 9 responses in 100, so this was never the game-changer it may have sounded like. For completeness, the other SMARCB1-directed routes are closed regardless: one trial runs only in France and requires complete loss of both gene copies, one US study has closed to enrolment, and two drug programmes were terminated by their companies.

The other half of the Rb stain. If the Rb stain shows the protein lost rather than kept, the Houston trial in Option 2 closes and a different early-stage trial at the same centre opens instead, testing a drug called REC-617. It has no efficacy data yet. The point is that one stain, already in the first step, routes you between the two.

The dose-ceiling trial. A study combining a drug called peposertib with a gentler form of doxorubicin is the one with the 360 ceiling that six full cycles of the planned regimen lands exactly on. It has no efficacy data yet either; it appears on this page as arithmetic, not as a promise.

Cell therapies your tissue type fits but your cancer does not. Several engineered-cell products, including one approved in a different sarcoma, require exactly the tissue type you have but are restricted to other cancer types or closed to enrolment. They are listed so the difference stays visible: the tissue type opened a door, and for these particular products the disease name closes it.

Negatives kept on purpose. Immune-checkpoint drugs on their own were tested in this disease and did not work, including a study in uterine leiomyosarcoma stopped early with zero responses. A strategy blocking a DNA-stress pathway called ATR was tested in this disease with the right biomarker selection, and the drug demonstrably hit its target, and patients still progressed within about four months. These entries are marked considered and not recommended, so the reasons are visible rather than silently missing.

Questions to ask your oncologist

- Before I sign the chemotherapy consent: the trial reported about 8 in 10 people had severe white-cell drops and about 28 in 100 were admitted with fever. What is our plan for that, do we use white-cell growth-factor injections from cycle one, and what number do I call at 2 a.m.?
- Can the tissue-release request go out this week as a single order naming both the 2024 uterine specimen and the 2026 lung biopsy, and who specifically owns chasing the outside hospital until the blocks arrive?
- Can we agree a doxorubicin stopping rule before cycle one, so the Toronto surgery study and the trial with the 360 ceiling stay open? What, if anything, do we give up by stopping at five cycles instead of six?
- If the INI1 stain comes back normal, which the published data say is likely, can we agree now that the

SMARCB1 finding is closed and not worth chasing further?
- Can my fitness score, heart scan, lung function, and oxygen level all be formally documented before the first cycle, with my home altitude written on the oximetry report?
- Should we contact the PRAME cell-therapy sponsor now to start screening in parallel, and ask them directly whether uterine leiomyosarcoma is accepted and whether they run their own marker test during screening?
- If the heart or lung workup rules out full-dose doxorubicin, what is the fallback, and does that fallback contain gemcitabine, which would permanently close the Houston trial?
- Nobody has actually recorded what I want: which side effects I would refuse, how far I would travel, what I am aiming for. Can we set aside time for that conversation before the first cycle?

Sources

PubMed (PMID): 9011700, 18534250, 21867989, 22595799, 24618336, 25713428, 26952295, 28440953, 28882536, 28988646, 29370992, 29891538, 29895706, 30541756, 32910151, 33035459, 34271067, 34303806, 34551229, 35835135, 37063124, 37143137, 37467452, 37532104, 38554725, 39231341, 39453756, 39478506, 39660994, 39918552, 40057343, 40152485, 40205198, 41285423, 41882006

ClinicalTrials.gov (NCT): NCT02601950, NCT02811523, NCT02997358, NCT03016819, NCT03686124, NCT03880019, NCT03965234, NCT03967223, NCT04044768, NCT04044859, NCT04262466, NCT04807816, NCT04939610, NCT04965753, NCT05286801, NCT05355753, NCT05420727, NCT05432791, NCT05691491, NCT05711615, NCT05999396, NCT06074692, NCT06088290, NCT06242470, NCT06277154, NCT06308419, NCT06402201, NCT06444880, NCT06498648, NCT06528769, NCT06551142, NCT06571734, NCT06710756, NCT06797999, NCT06849986, NCT06942442, NCT06981637, NCT07156565, NCT07303387, NCT07374848, NCT07516925, NCT07633756